

STATEMENT OF PURPOSE

Srashti Katiyar

B.Tech. (Computer Science – Artificial Intelligence) | Banasthali Vidyapith | CGPA: 9.19/10

Academic Foundation in Artificial Intelligence

I am currently in the third year of my B.Tech. degree in Computer Science and Artificial Intelligence. My academic foundation in the field of artificial intelligence is evident through my academic inclination towards machine learning, neural networks, and data-driven decision systems. My academic subjects include the basics of Machine Learning, Linear Algebra, Probability and Statistics, and the Design and Analysis of Algorithms.

These subjects have helped me gain significant knowledge regarding the understanding and derivation of the learning algorithms and the working mechanisms of the optimization functions. I am particularly inclined towards the mathematical understanding and intuition behind the learning systems.

Practical Exposure to Machine Learning

To gain further exposure to the understanding and application of machine learning algorithms, I have implemented the Perceptron and Multi-Layer Perceptron algorithms using the standard datasets such as the Iris and Breast Cancer datasets.

Through these implementations, I gained significant knowledge regarding the understanding and application of the training and testing performances and the generalization behaviour of the model. I gained significant exposure to the understanding and application of the critical factors such as overfitting and the variance and bias trade-offs.

Applied AI and Problem-Solving Orientation

Currently, I am working on a project titled “LifeLink – Every Second Counts.” This is aimed at improving the way medical funds are allocated. While working on this, I started getting interested in the ways that various features, such as priority prediction, anomaly detection, and verification, could be implemented using AI.

This has, therefore, given rise to an increased interest in developing an AI that not only has the capability to make decisions with the highest accuracy but also does so responsibly.

Research Interests

My research interests include machine learning, predictive modelling, optimization, and the deployment of various models. This is an area that has caught my attention, as I would like to know more about the ways that various models could be optimized for performance while also being able to explain them.

My main interest is structured learning methodologies, such as hypothesis formulation, experimental design, empirical evaluation, and performance benchmarking, as I would like to go beyond the implementation of the various algorithms and learn more about the analytical reasoning behind the various models that could be created using AI.

Why IIT Madras?

IIT Madras is a place where a highly stimulating environment is provided for the growth of intellect. The strong foundational background and impactful AI research at IIT Madras is a strong match for my academic aspirations.

This opportunity is a chance for me to work on live datasets, be a part of impactful research projects, and hone my skills under the guidance of experts. I am a highly motivated individual who is meticulous about my work and is highly passionate about enhancing my knowledge in the field of Artificial Intelligence.

I strongly believe that being a part of IIT Madras would greatly enhance my research potential as well as provide me with a broader perspective on the field.